Kavya Shah

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Education

University of California, San Diego *December 2026*

BS in Mathematics and Computer Science, Minor in Data Science *GPA: **3.89***

- ***Relevant Coursework:*** Machine Learning, Statistical NLP, Recommender Systems, Optimization, Database Systems, Probability & Statistics, Linear Algebra, Data Structures & Algorithms, Software Engineering

Skills

- ***Languages:*** Python, Java, C++, C, SQL, JavaScript, R, MATLAB.
- ***Frameworks & Libraries:*** PyTorch, TensorFlow, scikit-learn, LangChain, LangGraph, OpenCV, Pandas, React, Flask, FastAPI.
- ***Concepts:*** Agentic applications, Multi-agent systems, Deep Agents, RAG, Fine-tuning, Cloud AI Deployment, MCP.

Professional Experience

AI Engineering Intern, Qualcomm *June 2026- Sept 2026*

San Diego, CA

- Built an AI-native evaluation framework analyzing modem execution logs against expected behavior via event-driven failure detection.
- Architected parallelized **LangGraph** multi-agent system with routing across different base models– **65%** faster than simple tool-calling.
- Built **MCP** tools for test-step lookup and vector-embedding search over **Elasticsearch**, retrieving definitions only on uncertainty.
- Productionized services with **FastAPI**, **MongoDB**, **Docker**, **Kubernetes**, automating planned-vs-executed comparison and failure classification at scale.

AI Consultant, Pear.care *Sept 2025 – Jan 2026*

Remote

- Led development of **Deep Agents** orchestrating medical workflows (symptoms→ICD→CPT→doctors) through contextual routing.
- Integrated **MCP** to enable agents to securely access standardized research, calculator, and audit tools.
- Built scalable **RAG-based AWS pipelines** with multi-tenant security, ethical safeguards, and task scheduling for clinical AI.
- Developed **semantic search pipelines** using embeddings and LLMs to interpret medical intent and retrieve evidence.

Database Engineer, San Diego Supercomputer Center (SDSC) *Aug 2025 – Present*

San Diego, CA

- Engineered secure **data migration pipelines** extracting congenital heart disease datasets from **AWS DynamoDB** using **Boto3**.
- Transformed semi-structured clinical and MRI metadata into standardized **JSON schemas** and migrated data into **PostgreSQL**.
- Built reproducible **PostgreSQL dumps** and **Dockerized pipelines** enabling scalable analytics and multi-center research access.

Project Manager, Skin Lesion Detection (CSE Society @UCSD) *2024 – Present*

San Diego, CA

- Led design of a **melanoma detection model** using **ResNet-50 CNNs**, transfer learning, and cross-validation techniques.
- Developed **image preprocessing pipelines** with resizing, normalization, and augmentation to improve deep model robustness.
- Implemented **TensorFlow** and **PyTorch** workflows for model quantization, compression, and efficient on-device inference.
- Achieved 91.3% accuracy, 0.89 ROC-AUC, and presented research findings at **SAIRS 2025 Conference**.

Research Assistant - Bioinformatics & Data Analytics, UC San Diego Health *2025 – Present*

San Diego, CA

- Designed **ML pipelines for scRNA-seq**, integrated with ATAC-seq and RNA velocity for multimodal immune profiling.
- Built Scanpy & scVI pipelines integrating scRNA-seq & ATAC-seq; deployed reproducible **multimodal workflows**.
- Discovered **tumor-specific immune cell clusters** post-immunotherapy, providing potential biomarkers for treatment response.

Projects

NeuroMap *(Python, FastAPI, scikit-learn, HTML/JS)* *2026*

- Predicts cognitive flexibility and academic performance from daily habits using three ML models (Random Forest, Gradient Boosting, K-Means) trained on 212 students across 43 engineered features.
- Surfaces key performance drivers, tiered personalized advice, and a real-time habit simulator via FastAPI backend.

PrepAI – AI Interview Coach *(Python, FastAPI, JavaScript, Gemini API)* *2026*

- Converts any job description into a **personalized mock interview**: six role-specific questions, live AI feedback per answer, and a full performance report with scoring and hire-likelihood estimates.
- Generates filler-word analysis and tailored action plans using **Gemini-powered evaluation** pipeline.

TumorSense-Data Science Student Society *(Python, scikit-learn, SVM, NumPy) – Spring Cohort 2026* *2026*

- End-to-end **SVM classification pipeline** on the Wisconsin Breast Cancer Diagnostic dataset; reduces 30 cell-nucleus measurements to the 10 most predictive features.
- Achieved **99.8% ROC-AUC and 97% F1 score**; deployed as a live web demo for instant benign/malignant predictions.

Travelog – San Diego Supercomputer Center *(Java, C++, React, Node.js, Flask, MongoDB, AWS, Next.js)* *2025*

- Engineered AI-assisted travel planning features using React, Flask APIs, and integrated user preference learning models.
- Deployed AWS CI/CD pipelines that reduced UI–API deployment bugs by 30% and improved production stability overall.